Status	Finished
Started	Wednesday, 21 October 2026, 10:40 PM
Completed	Wednesday, 21 October 2026, 10:44 PM
Duration	4 mins 8 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 94 and 99

b. 90 and 99 ✓

c. 90 and 100

d. 89 and 94

Question **2**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search has a lower time complexity on large, sorted lists ✓

b. Binary search can find multiple instances of the target element

c. Binary search works on unsorted lists

d. Binary search does not require dividing the list

Question **3**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is not the array at all

b. When the item is the last element in the array

c. Item is the last element in the array or item is not there at all

d. When the item is somewhere in the middle of the array ✓

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

- a. It requires the list to be sorted ✓
- b. It can only be applied to large lists
- c. It does not work with negative numbers
- d. It is slower than linear search for small lists

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

- a. Binary search
- b. Linear search ✓
- c. Hashing
- d. Dynamic search

Question **6**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

- a. Both (1) & (3)
- b. Linear
- c. Hash
- d. Binary ✓

Question **7**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

- a. By dividing the list into halves
- b. By comparing each element in the list sequentially ✓
- c. By using a hash function
- d. By sorting the list first

Question **8**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n/2
b. n ✓
c. n-1
d. log n

Question **9**

Correct

Mark 1.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The higher middle element is chosen as the middle element ✓
b. The middle element is chosen randomly
c. The search stops
d. The lower middle element is chosen as the middle element

Question **10**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. At the beginning of the list
b. In the middle of the list
c. In the left sublist
d. In the right sublist ✓

Question **11**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search backtracks to find duplicate elements
b. The search starts over from the beginning
c. The search continues until the end of the list
d. The search stops immediately ✓

Question **12**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

- a. Determining whether an element is present in a list ✓
- b. Deleting elements from a list
- c. Inserting elements into a list
- d. Sorting elements in a list

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

- a. Checking each element sequentially ✓
- b. Skipping every second element
- c. Dividing the list in half repeatedly
- d. Sorting the list before searching

Question **14**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

- a. Linear search ✓
- b. Hashing
- c. Binary search
- d. Jump search

Question **15**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

- a. 100
- b. 10 ✓
- c. 500
- d. 1000